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Jupiter's ionosphere: Results from the first Galileo radio occultation experiment

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Abstract. The Galileo spacecraft passed behind Jupiter on December 8, 1995, allowing the first radio occultation measurements of its ionospheric structure in 16 years. At ingress (24°S, 68°W), the principal peak of electron density is located at an altitude of 900 km above the 1-bar pressure level, with a peak density of 10^5 cm^{-3} and a thickness of ~200 km. At egress (43°S, 28°W), the main peak is centered near 2000 km altitude, with a peak density of 2×10^4 cm^{-3} and a thickness of ~1000 km. Two thin layers, possibly forced by upwardly propagating gravity waves, appear at lower altitudes in the ingress profile. This is the first in a two-year series of observations that should help to resolve long-standing questions about Jupiter's ionosphere.

Introduction

As the first spacecraft to orbit Jupiter, Galileo will perform 22 radio occultation experiments at Jupiter (8), Io (6), Europa (3), Ganymede (4), and Callisto (1) during its two-year primary mission. This paper reports results from the first occultation of Galileo by Jupiter.

These experiments cannot be conducted as originally planned [*Howard et al.* 1992] because the spacecraft high-gain antenna failed to deploy. A low-gain antenna (LGA) must be used instead, with two unfortunate consequences. First, the LGA can transmit only one signal, instead of two signals at coherently-related frequencies. Second, the intensity of the signal received on Earth is greatly reduced, resulting in a signal-to-noise ratio (SNR) of ~15 dB in a 1-Hz bandwidth. Measurements of Jupiter's neutral atmosphere, which causes substantial defocusing, are extremely difficult at this modest SNR. Hence, this paper addresses only the retrievals in Jupiter's ionosphere, which provide new insight into a region distinguished by its complex heterogeneity.

Procedure and Results

On December 8, 1995, the Galileo spacecraft disappeared behind Jupiter for 3.7 hours. During the 6.2 hours centered on the occultation, the spacecraft LGA radiated a coherent signal at a frequency of 2.3 GHz derived from an ultrastable quartz oscillator (USO) on board. This signal was tracked by the 70-m diameter antenna of NASA's Deep Space Network near Madrid, Spain. The received signal was mixed to audio frequencies, filtered, digitized to 8-bit samples with a 5-kHz sampling rate, and recorded on magnetic tape.

We extracted the time history of signal frequency through Fourier analysis of these data. We then obtained residual frequencies by subtracting the frequency variation that would have been observed in the absence of an atmosphere/ionosphere on Jupiter. These residual frequencies exhibit a small long-term drift, of order 10^{-4} Hz sec^{-1}, presumably from instability of the USO and refraction in the interplanetary plasma and Earth's ionosphere. We removed this drift through use of a simple function fitted to the frequency residuals over a baseline interval well above Jupiter's ionosphere. Separate corrections were applied at ingress and egress. Figure 1 displays the results at ingress.

We retrieved vertical profiles of refractive index, n, from the frequency residuals using an established pro-

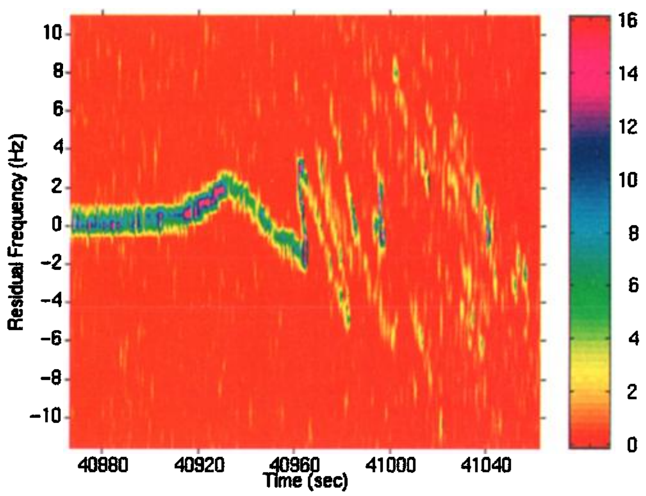

Figure 1. Spectral power vs residual frequency and time at occultation entry. Time span includes principal effects of Jupiter's ionosphere. The figure is constructed from a series of power spectra, each computed from 1.6 sec of data. Power in each frequency bin is color coded and each spectrum is displayed as a vertical strip. Power scale is linear with arbitrary units. Times correspond to Earth reception in sec past midnight (UTC) on December 8, 1995.

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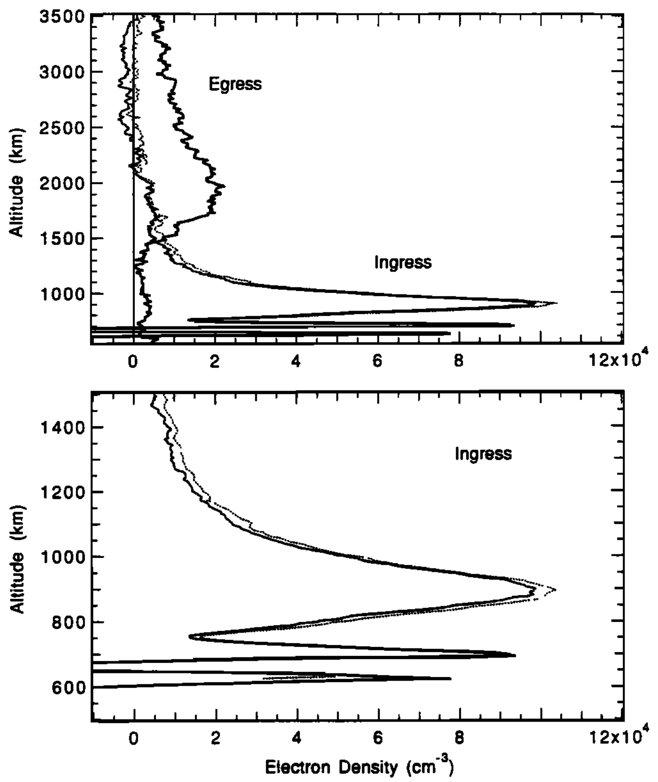

Figure 2. Profiles of electron density in Jupiter's ionosphere. Upper panel compares profiles at ingress and egress. Lower panel shows ingress profile in the vicinity of the main peak. Dashed and solid lines show results from different methods of analysis. Altitudes are above the 1-bar geopotential [*Lindal et al.* 1981].

cedure [e.g., *Fjeldbo et al.* 1971]. This requires detailed knowledge of the experiment geometry, which was provided by the Galileo Navigation Team. Within the ionosphere, the refractive index at 2.3 GHz is due predominantly to free electrons and the electron density is proportional to $n - 1$. These steps yielded the electron density profiles in Fig. 2.

There is a simple correspondence between structures in Figs. 1 and 2. For example, the feature in the ingress timeline at 40910-40960 sec, consisting of a rise in residual frequency followed by a decrease, is the signature of the electron density peak centered near 900 km. This peak has a maximum density of 10^5 cm^{-3} and a thickness of ~200 km. The characteristics of the main peak at egress are considerably different. It is higher, near 2000 km altitude, broader, with a thickness of ~1000 km, and weaker, having a peak density of 2×10^4 cm^{-3}.

At ingress, Jupiter's ionosphere contains two thin layers at lower altitudes, which produced significant refractive bending of radio signals that passed tangent to their flanks. The deflection is away from the altitude of peak density. This allowed radio signals to propagate from Galileo to Earth along a variety of paths through the ionosphere, deflecting upward or downward in the vicinity of the sharp layers, with several distinct signals arriving simultaneously at the receiving antenna. Each signal had a different Doppler-shifted frequency depending on its angle of departure from the spacecraft. This phenomenon—multipath propagation—is responsible for the striking pattern of "zig-zags" slanting from upper left to lower right beyond 40960 sec in Fig. 1.

The multivalued frequency residuals are deciphered by noting that altitude changes monotonically as one traces along the locus of received signal in Fig. 1. For example, the undulation of the curve beginning near 40964 sec, increasing in frequency while moving backward in time, followed by a decrease in frequency while moving forward in time to a minimum near 40982 sec, corresponds to a monotonic decrease in altitude from 760 to 680 km, encompassing the upper thin layer.

Figure 2 compares profiles at ingress obtained by two methods. In the first (solid line), signal frequencies were estimated from spectra like those in Fig. 1 and a profile was retrieved as described above. However, the presence of multiple signals, some of which are very weak and whose total number is unknown a priori, presents a significant challenge to signal detection and frequency estimation. This led us to try a second method (dashed line), which uses a "Fresnel filter" to remove multipath, defocusing, and diffraction effects from the data [*Marouf et al.* 1986; *Karayel and Hinson* 1997]. In theory, this step will yield a signal whose frequency is single valued and whose amplitude is nearly uniform. Profile retrieval can then proceed as described above, but with complications due to multipath and strong defocusing largely eliminated. Results from the two approaches agree closely, exhibiting a significant difference only near the lower thin layer, where retrieval is difficult because of low SNR.

There is a suggestion in Fig. 1 of additional thin layers at lower altitudes, beyond 40990 sec, but the sporadic nature of these signals prevents retrieval of electron densities at lower altitudes. Likewise, narrow layers in the lower ionosphere produced multipath during egress, but we have not yet been able to resolve any thin layers.

Table 1 summarizes the experiment geometry. The latitudes and longitudes correspond to the altitude of maximum density in each profile. Magnetic field angles were calculated using the O_4 model [*Acuña et al.* 1983].

Table 1. Occultation Geometry

	Ingress	Egress
Range to Jupiter (10^3 km)	855	1034
Latitude (planetocentric)	24°S	43°S
Longitude (System III, 1965)	68°W	28°W
Magnetic Field Angles		
Dip	58°	55°
Declination	35°W	4°E
Solar Zenith Angle	89°	91°
	evening	morning

Discussion

Previous radio occultation experiments with Pioneers 10 and 11 [*Fjeldbo et al.* 1975, 1976] and Voyagers 1 and 2 [*Eshleman et al.* 1979a, b] yielded a total of 7 profiles of electron density in Jupiter's ionosphere. With so little data available, it is not surpising that fundamental questions about the ionosphere remain unanswered.

Simple photochemical models cannot reconcile the major features of the electron density profiles measured by Voyager with the neutral atmospheric structure deduced from Voyager Ultraviolet Spectrometer (UVS) data [e.g., *McConnell et al.* 1982; *Waite and Cravens* 1987]. The predicted peak density is too large by nearly an order of magnitude, while the predicted altitude of the peak is too low by as much as 1000 km. *McConnell et al.* and *Majeed and McConnell* [1991] offered viable explanations for both problems by incorporating two additional effects into their ionospheric models. (See also *Strobel and Atreya* [1983].) Adopting an idea proposed by *McElroy* [1973], they demonstrated that the observed peak densities can be explained through vibrational excitation of H_2, which provides a rapid loss mechanism for otherwise long-lived H^+. Calculations by *Cravens* [1987] and *Majeed et al.* [1991] indicate that such elevated vibrational temperatures are likely to be present on Jupiter. At the same time, *McConnell et al.* and *Majeed and McConnell* showed that vertical drift, forced by either electric fields or meridional winds, can account for the altitude discrepancy. Comparisons between measurements and model predictions imply that the vibrational temperature and vertical drift speed vary dramatically with location on Jupiter.

In a complementary study of Jupiter's thermosphere and ionosphere, *Waite et al.* [1983] stressed the importance of particle precipitation, which *McConnell et al.* and *Majeed and McConnell* did not include in their models. *Waite et al.* showed that the thermospheric energy budget is strongly influenced by electron precipitation, especially at high latitudes, where intense auroral heating will drive a meridional circulation at ionospheric heights that redistributes heat globally. (See also [*Waite et al.* 1997].) Their calculations demonstrate that electron precipitation can influence ionospheric structure in several important ways. One is through direct ionization, which can produce electron densities exceeding 10^7 cm^{-3} at high latitudes. Indirect effects arise through vibrational heating of H_2 and the aurorally-driven meridional circulation [*Strobel and Atreya* 1983], the two mechanisms that have been proposed to explain the Voyager electron density profiles. In this regard, the structure of Jupiter's ionosphere appears to depend strongly on particle precipitation.

Against this background, we now consider the major features of the Galileo profiles. Simple photochemical reactions will produce an ionospheric peak at an altitude where the number density of H_2 is about 10^{11} cm^{-3} [*McConnell et al.* 1982]. According to Galileo probe

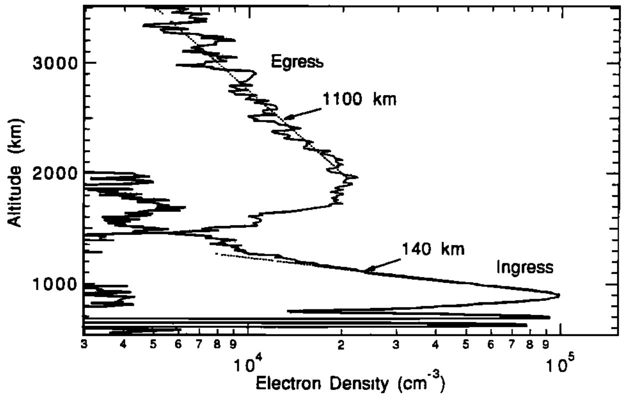

Figure 3. Profiles from Fig. 2 with density on a logarithmic scale.

measurements, this occurs at about 700 km [*Seiff et al.* 1997]. The profiles in Fig. 2 have peaks at altitudes of about 900 and 2000 km at entry and exit, respectively. Thus, the former is roughly consistent with a simple photochemical model, while the latter is considerably higher than predicted, as was the case in three of the four electron density profiles obtained with Voyager. As for the peak densities, the results at Galileo entry and exit differ by a factor of 5, comparable to the variations among Voyager profiles.

Turning to the topside structure (Fig. 3), the Galileo egress profile has a decay scale of 1100 km, similar to the 880-km decay scale measured during egress of Voyager 2 at 51°S. At these altitudes (>2000 km), the principal ion is H^+ so that a decay scale of 1100 km implies an average temperature of 1500 K for ions and electrons in diffusive equilibrium. Because the plasma temperature should exceed the neutral gas temperature, this is consistent with Voyager UVS solar occultation measurements, which indicate a neutral temperature of 1100±200 K near this altitude [*Atreya et al.* 1981]. Topside structure at ingress is more difficult to interpret, in part because the density of H_3^+ may be substantial at altitudes below 1500 km [*Majeed and McConnell* 1991].

The Galileo ingress profile contains two thin layers separated by ~70 km at altitudes between 600 and 750 km. This structure is reminiscent of the multiple, narrow layers of ionization observed at similar altitudes in radio occultations of Pioneers 10 and 11. Some of these may arise from an external source of heavy ions [e.g., *Chen* 1981], but this interpretation cannot readily account for the presence of multiple layers. Model calculations by *Kim and Fox* [1994] indicate that multiple, narrow layers of hydrocarbon ions are likely to be present at altitudes of 300-450 km. Scattering and multipath caused by such layers may be responsible for our inability to retrieve profiles at altitudes below ~500 km. However, the layers observed at 600-750 km are too high in altitude to be attributed to hydrocarbons.

Alternatively, the thin layers in Fig. 2 could be forced by upwardly propagating gravity waves, which account for much of the sporadic E-layer activity in Earth's ionosphere at altitudes 130-200 km [e.g., *Kelley* 1989]. In the simplest case, the ions and electrons tend to follow the dominant neutral gas oscillating transverse to the direction of gravity-wave propagation. However, the charged particles are bound to the magnetic field, and their motion is the neutral motion projected along the field lines. In general, the ion-electron motion is therefore not transverse, and layers of compression and rarefaction form. For the ions to be bound to field lines, the ion gyrofrequency must be large compared to the ion-neutral collision frequency. In Jupiter's upper atmosphere this occurs above ~400 km, near the homopause. At high altitudes, gravity waves will be dissipated by molecular viscosity, but waves with a vertical wavelength of ~70 km can persist to 700 km or higher, which marks the topmost narrow peak in Fig. 2. Thus, gravity-wave forcing seems to be consistent with the observed structure of the thin layers.

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